

EM exercise 04.m

```

1  clc;
2  clear;
3  close;
4
5  c = -8i;
6  p0 = abs(c);
7  ang0 = arg(c);
8
9  p1 = p0^(1/3);
10 p2 = p1;
11 p3 = p1;
12
13 ang1 = (ang0/3);
14 ang2 = ang1+(2*pi/3);
15 ang3 = ang1+2*(2*pi/3);
16
17 x1 = p1*cos(ang1);
18 y1 = p1*sin(ang1);
19
20 x2 = p2*cos(ang2);
21 y2 = p2*sin(ang2);
22
23 x3 = p3*cos(ang3);
24 y3 = p3*sin(ang3);
25
26 figure
27 plot([0, x1], [0,y1], [0, x2], [0,y2], [0, x3], [0,y3]);
28 xlabel("Real");
29 ylabel("Imaginary");
30 title("Roots of Complex Number 0-8i");
31 grid on;
32 axis([-3, 3, -2.5, 2.5]);
33

```

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33
34 text(x1,y1,["(" num2str(p1) ", " num2str(ang1) ")"], "fontsize",15);
35 text(x2,y2,["(" num2str(p2) ", " num2str(ang2) ")"], "fontsize",15);
36 text(x3,y3,["(" num2str(p3) ", " num2str(ang3) ")"], "fontsize",15);
37
38 xx = linspace(-p1, p1, 50);
39 yy1 = sqrt(p1^2 -xx.^2);
40 yy2 = -1*sqrt(p1^2 -xx.^2);
41 hold on;
42 plot(xx,yy1,"linestyle","--",'c',xx,yy2,"linestyle","--",'c');
43
44 k1 = p1*exp(i*ang1);
45 k2 = p2*exp(i*ang2);
46 k3 = p3*exp(i*ang3);
47
48

```

Figure 1

File Edit Tools



Roots of Complex Number 0-8i

